Week 24: Three random decks

Adult guide built from the problem files · Unpiloted. Physical pretests unperformed.

This week's adult guide is week-24-facilitator.pdf, whose source is editable/facilitator-src/build.py. These problem files hold each problem's one-sentence idea, not its answers; the answers, hints and proofs are in the guide.

Problems and routes

Problem	K–1	2–3	4–5	route
Play A against B, B against C and C against A	P1			first
A against B, all nine pairs	P2			first
B against C, all nine pairs	P3			first
C against A, and the cycle	P4			first
Choose the deck with the advantage and play	P5			return
Change A's largest card	P6			return
One swap between A and B	P7			return
Two decks from 1 to 6 that tie	P8			return
Pick first or pick second		P1	P1	first
Compare A, B and C		P2	P2	first
Guess the hidden decks from six results		P3		first
Keep 9, 8 and 7; place 1 to 6		P4	P3	first/return
Choose A's middle card		P5		return
Two copies of every card, or of some		P6		return
A cycle with totals 15, 18 and 21		P7		return
A cycle with repeated values			P4	return
Six-card and four-card chances			P5	return
A rule for two-card decks			P6	return
Three two-card decks from 1 to 6	P9	P8		return
The fewest cards per deck for a cycle			P7	return
Can every advantage reach six of nine?			P8	return

First route. K–1 P1–P4; Grades 2–3 P1–P4; Grades 4–5 P1–P3. **Return visits.** K–1 P5–P9; Grades 2–3 P4–P8; Grades 4–5 P3–P8.

Key, one entry per problem

Play A against B, B against C and C against A

K–1 P1 · first route

Idea. Each game is a short random experiment: in each matchup the deck with five of nine winning pairs wins a round with chance 5/9, so six rounds often, but not always, favour it, and no deck is ahead in both of its games.

A against B, all nine pairs

K–1 P2 · first route

Idea. Of the nine equally likely card pairs, A wins five: (2,1), (4,1), (9,1), (9,6) and (9,8).

B against C, all nine pairs

K–1 P3 · first route

Idea. Of the nine card pairs, B wins five: (6,3), (6,5), (8,3), (8,5) and (8,7).

C against A, and the cycle

K–1 P4 · first route

Idea. C wins five of nine pairs against A, so A beats B, B beats C and C beats A: no deck beats both others.

Choose the deck with the advantage and play

K–1 P5 · return route

Idea. Against A choose C, against B choose A, against C choose B; each chosen deck wins a round with chance 5/9, but it need not win more of six rounds.

Change A's largest card

K–1 P6 · return route

Idea. With A = (2,4,x), A wins 2 + (number of B cards below x) pairs, so of 3, 5, 7 and 9 only 9 gives A more than half.

One swap between A and B

K–1 P7 · return route

Idea. Exactly the five swaps in which A gives away a card bigger than the one it receives make B win: (2,1), (4,1), (9,1), (9,6), (9,8).

Two decks from 1 to 6 that tie

K–1 P8 · return route

Idea. Impossible: two three-card decks give nine pairs, each with one winner, and nine cannot be split equally.

Pick first or pick second

Grades 2–3 P1 · Grades 4–5 P1 · first route · wording differs by band

Idea. Whatever deck the first player picks, the second can pick the deck that beats it 5 pairs to 4, so picking second is the advantage; ten rounds show this only as a tendency.

Compare A, B and C

Grades 2–3 P2 · Grades 4–5 P2 · first route · wording differs by band

Idea. A beats B, B beats C and C beats A, each 5 pairs to 4 (chance 5/9); no deck beats both others, and every deck totals 15, so totals do not decide.

Guess the hidden decks from six results

Grades 2–3 P3 · first route

Idea. Every finite run of results can happen whichever bag has the advantage, so no fixed number of draws makes it certain.

Keep 9, 8 and 7; place 1 to 6

Grades 2–3 P4 · Grades 4–5 P3 · first/return route · wording differs by band

Idea. Exactly three arrangements make the cycle: the decks A, B, C, (1,5,9), (3,4,8), (2,6,7), and (2,3,9), (1,6,8), (4,5,7).

Choose A's middle card

Grades 2–3 P5 · return route

Idea. With A = (2,x,9), only x = 2 or x = 4 keeps all three 5-to-4 advantages; 0 loses A over B, and 9 or 10 loses C over A.

Two copies of every card, or of some

Grades 2–3 P6 · return route

Idea. Doubling every card multiplies every pair count by four (A over B 20 of 36), so no winner changes; the four-card decks (2,2,4,9), (1,1,6,8), (3,3,5,7) give A over B 10 of 16 and C over A 10 of 16, but C now beats B 9 to 7.

A cycle with totals 15, 18 and 21

Grades 2–3 P7 · return route

Idea. Exactly one choice works: A = (3,5,7), B = (2,4,12), C = (1,9,11); the smallest total can still win its comparison.

A cycle with repeated values

Grades 4–5 P4 · return route

Idea. Repeated values allow cycles such as A = (1,4,4), B = (3,3,3), C = (2,2,5), winning 6, 6 and 5 of 9.

Six-card and four-card chances

Grades 4–5 P5 · return route

Idea. Doubling every card keeps A's chance against B at 20/36 = 5/9; the four-card decks give A over B 10/16, B over C 7/16 and C over A 10/16, so B no longer beats C.

A rule for two-card decks

Grades 4–5 P6 · return route

Idea. A two-card deck wins more than half against another exactly when its smaller card beats the smaller and its larger card beats the larger; a split gives a 2-2 tie.

Three two-card decks from 1 to 6

K–1 P9 · Grades 2–3 P8 · return route · wording differs by band

Idea. Impossible: two-card decks with no shared number win by beating smaller with smaller and larger with larger, which cannot go round a cycle.

The fewest cards per deck for a cycle

Grades 4–5 P7 · return route

Idea. One-card and two-card decks cannot cycle (numbers, and two-card dominance, are transitive), and A, B, C show three cards suffice, so three is the fewest.

Can every advantage reach six of nine?

Grades 4–5 P8 · return route

Idea. Impossible: no split of 1 to 9 into three decks of three gives each deck at least six of nine pairs against the next.